MAKE THINGS HAPPY !

4x AI-ADS1115 - IoTextra Analog

Analog Input Module

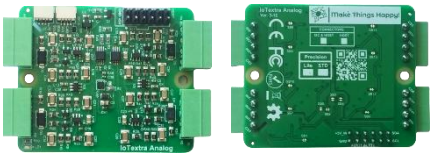


The **4x AI-ADS1115 - IoTextra Analog** module (hereinafter **IoTextra Analog**, version 3.12) is an analog input module based on the 16-bit ADS1115 ADC from Texas Instruments. It is compatible with most types of microcontrollers (the **I²C** interface is used for communication).

The module uses operational amplifiers and passive components with low error to reduce noise influence and ensure accuracy corresponding to the ADC resolution.

Input power: +5VDC. It includes reverse polarity protection.

The module allows measuring four differential analog signals, both voltage and current. The two ADCs installed on the module have a 16-bit resolution. The ADC type is ΔΣ (delta-sigma).
Voltage measurement ranges: 0-0.5V, 0-5V, 0-10V, ±0.5V, ±5V, ±10V.
Current measurement ranges: 0-20mA, 4-20mA, ±20mA, 0-40mA.
Measurement speed: up to 860 samples per second for the **ADS1115**.



APPLICATIONS:

🏭 **Industrial Process Monitoring**
Measures 4-20mA transmitter outputs from pressure, temperature, and flow sensors, or reads 0-10V analog signals directly — no external signal conditioning needed thanks to the onboard PGA and precision voltage reference.
Recommended software: Node-RED, IoTflow

🤖 **Robotics & Lab Automation**
Captures voltage and current outputs from bench power supplies, sensor calibration rigs, and test equipment across four independent differential channels — 16-bit resolution and selectable gain suit precision lab measurements.
Recommended software: Node-RED, IoTflow

🌱 **Smart Farming**
Reads 4-20mA soil moisture, pH, or EC sensors alongside 0-10V feedback from irrigation valve actuators — differential inputs reject common-mode noise from long field wiring runs.
Recommended software: Node-RED, IoTflow

🏠 **Smart Home & Consumer Electronics**
Monitors battery charger voltage/current curves, appliance sensor feedback, or infotainment system analog inputs — configurable per-channel voltage or current range covers most consumer sensor interfaces.
Recommended software: Node-RED (MQTT) + Blynk

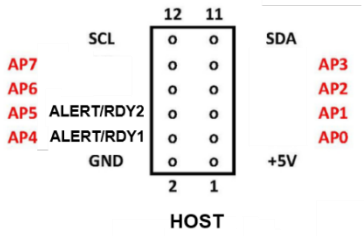
FEATURES:

- Compatibility with major known microcontrollers
- Module power supply is 5VDC
- Protection against reverse power supply polarity is included
- ADC resolution - 16 bits (15 bits effective for unipolar ranges)
- ADC type - ΔΣ (delta-sigma)
- 4 differential analog input channels
- Voltage measurement ranges:
  0-0.5V, 0-5V, 0-10V,
  ±0.5V, ±5V, ±10V

- Current measurement ranges:
  0-20mA, 4-20mA, ±20mA, 0-40mA
- ADC measurement speed up to 860 samples per second per ADC (each ADC serves two channels, so up to approx. 430 samples per second per channel when both channels are scanned)
- **I²C** interface: 100kHz, 400kHz, 3.4MHz (if supported by the host)
- Connection to the module via the **I²C** bus is done through **Qwiic©** connectors or through pins 11 (**SDA**) and 12 (**SCL**) of the **HOST-P12** connector
- Transient suppression and electrostatic discharge protection of signals on **Qwiic©** connectors is provided by a TVS diode assembly (ESD protection)
- The **I²C** addresses of the ADCs are changeable: the ADC for channels 1 and 2 has an address of 1001001x (which can be changed to 1001000x using **SB2** on the reverse side of the module), and the ADC for channels 3 and 4 has an address of 1001011x (which can be changed to 1001010x using **SB3** and **SB4** on the reverse side of the module)
- Programmable Gain Amplifier (PGA)
- Module size: 47x56 mm. The module has mounting holes that allow it to be installed on a base module or a Raspberry Pi.

MODULE USAGE AND HOST CONNECTOR

A 12-pin **HOST** connector is always installed on the **IoTextra Analog** module. The connector pins are shown in the following figure



The module can be used in the following ways:

1) ***Standalone Use***: In this mode, the input signal states are available through the **HOST** connector or via the **Qwiic©** connectors on the **I²C** bus
2) ***Smart Use***. In this mode, an **IoTsmart** module with a microcontroller is plugged into the **HOST** connector. Communication with the **IoTextra Analog** module is through the **HOST** connector. Power is drawn from the **IoTsmart** module. The figure below shows the **IoTextra Analog** module with an **IoTsmart** module:



IoTextra Analog with **IoTsmart ESP32-S3**



IoTextra Analog with **IoTsmart RP2040**



IoTextra Analog with **IoTsmart RP2350A**

3) ***Mezzanine Use***. In this mode, the **IoTextra Analog** module is installed in a base module, and communication with it is through the **HOST-P12** connector. The figures below show the **IoTextra Analog** module installed in **IoTbase** modules:



IoTextra Analog and **IoTbase PICO** with **PICO2**



IoTextra Analog and **IoTbase Nano** with **ESP32-S3-Nano**



IoTextra Analog and **IoTbase Nano** with **STM Nucleo-32**

VOLTAGE AND CURRENT MEASUREMENT CALCULATION

The measurement circuit for each channel consists of an operational amplifier, the Programmable Gain Amplifier (PGA) built into the **ADS1115**, and the 16-bit ADC (delta-sigma type) itself. The gain of the operational amplifier depends on the resistance **R**, which can be 24.95kΩ or 49.9kΩ (determined by a pair of jumpers for each channel), and is 24.95kΩ by default. 49.9kΩ improves accuracy only for the 0-0.5V and ±0.5V ranges; for 0-10V, ±10V and 0-40mA it is not allowed (the op-amp output would exceed its usable range).

The output voltage of the operational amplifier, depending on the measured signal range and considering the **+2.5V** reference voltage, will be as follows (the usable output range of the operational amplifier is approx. 0.05…4.95V; with R=49.9kΩ, the ±5V and ±20mA ranges operate close to its lower limit). The ADC measures the differential voltage between the operational amplifier output and the 2.5V reference, i.e. (R/105kΩ)×Vin, therefore the recommended PGA is selected for this differential voltage. In the Recommended PGA column, the left value applies to R=24.95kΩ and the right value to R=49.9kΩ (for the 0.5V ranges with R=49.9kΩ, the signal uses about 93% of the ±0.256V scale; use PGA 8 if over-range detection is required):

Range	Input Signal		Op-Amp Output Signal		Recommended PGA	
	I_{in}	V_{in}	V for R=24.95kΩ	V for R=49.9kΩ		
0-0.5V		0	2.5	2.5	16	16
		0.5	2.618809524	2.737619048		
±0.5V		-0.5	2.381190476	2.262380952	16	16
		0.5	2.618809524	2.737619048		
0-5V		0	2.5	2.5	2	1
		5	3.688095238	4.876190476		
±5V		-5	1.311904762	0.123809524	2	1
		5	3.688095238	4.876190476		
0-10V		0	2.5	n/a	1	n/a
		10	4.876190476	n/a		
±10V		-10	0.123809524	n/a	1	n/a
		10	4.876190476	n/a		
0-20mA	0	0	2.5	2.5	2	1
	0.02	4.98	3.683342857	4.866685714		
±20mA	-0.02	-4.98	1.316657143	0.133314286	2	1
	0.02	4.98	3.683342857	4.866685714		
4-20mA	0.004	0.996	2.736668571	2.973337143	2	1
	0.02	4.98	3.683342857	4.866685714		
0-40mA	0	0	2.5	n/a	1	n/a
	0.04	9.96	4.866685714	n/a		

The Full-Scale Range (FS) depends on the gain coefficient (PGA)

PGA SETTING	FS (V)
2/3	±6.144V(1)
1	±4.096V(1)
2	±2.048V
4	±1.024V
8	±0.512V
16	±0.256V

(1) This parameter expresses the full-scale range of the ADC scaling. In no event should more than VDD + 0.3V be applied to this device.

To determine the **measured voltage**, the following equation should be used:

$$\text{ADCNumber} = 2^{15} \times \frac{1}{FSR} \times \frac{R}{105k\Omega} \times V_{in}$$

where:

R - is the resistance, which can take the value 24.95kΩ or 49.9kΩ depending on the jumpers
FSR - is the positive full-scale range, determined by the range setting in the PGA[2:0] configuration register
$\mathbf{V_{in}}$ - is the input voltage

To determine the **measured current**, the following equation should be used:

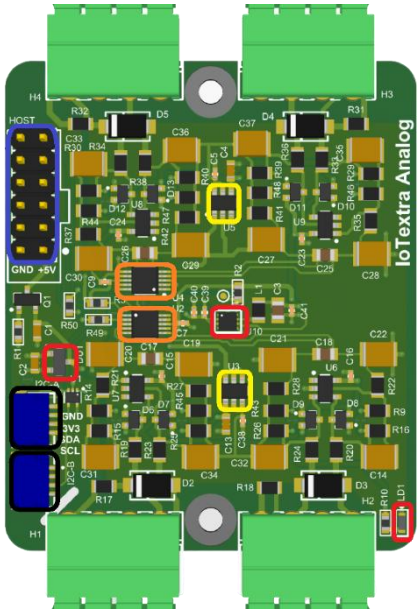
$$\text{ADCNumber} = 2^{15} \times \frac{1}{FSR} \times \frac{R}{105k\Omega} \times 249\Omega \times I_{in}$$

where:

R - is the resistance, which can take the value 24.95kΩ or 49.9kΩ depending on the jumpers
FSR - is the positive full-scale range, determined by the range setting in the PGA[2:0] configuration register
$\mathbf{I_{in}}$ - is the input current (effective shunt ≈248.7Ω: 249Ω loaded by the 210kΩ amplifier input)

COMPONENT LAYOUT

The figure below shows the component placement on the **top-side** of the **IoTextra Analog** module:



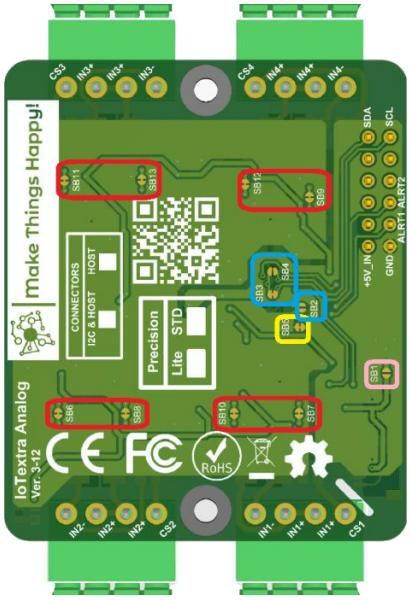
The main **HOST** connector is highlighted in blue.
The **Qwiic©** connectors are highlighted in black.
Highlighted in red are:

- The high-precision DC-DC converter **TPS62173**, used to obtain the analog +5V power
- The low-noise LDO used to obtain +3.3V
- The analog power indicator

The **ADS1115** ADCs are highlighted in orange.
The high-precision 2.5V reference voltage sources (**REF3425** type) are highlighted in yellow.

JUMPERS

The figure below shows the component placement on the **bottom-side** of the **IoTextra Analog** module:



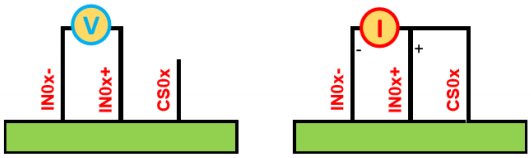
The following jumpers are located on the **bottom-side** of the module:

- **SB1** (highlighted in pink) - for connecting +3V3 within the module and the **Qwiic**© connectors. By default, the jumper is open, meaning there is no +3V3 power connection.
- **SB2** (highlighted in light blue) - for selecting the **I²C** interface address of the ADC for analog input channels 1 and 2 (this ADC is **U2** in the schematic). By default, this jumper is open. This means the ADC address is 1001001x. If this jumper is closed, the address of this ADC becomes 1001000x.
- **SB3** and **SB4** (highlighted in light blue) - for selecting the ADC address for analog input channels 3 and 4 (**U4** in the schematic). By default, jumper **SB3** is closed and SB4 is open. This means the address is 1001011x. If jumper **SB3** is opened and jumper **SB4** is closed, the address becomes 1001010x. Other states for jumpers **SB3** and **SB4** are not permissible.
- **SB5** (highlighted in light yellow) - connects the digital **GND** and analog **AGND** planes at a single point. Keeping analog and digital grounds separate elsewhere on the board prevents digital switching noise from coupling into the sensitive analog measurement circuits; SB5 provides the one intentional tie-point between them. By default, the jumper is **closed** — this is the correct setting for standalone use, ensuring the analog ground has a return path. Open this jumper only if the module is used alongside other boards where AGND is already tied to GND elsewhere in the system (to avoid a ground loop).
- For selecting resistances, the jumpers are: **SB7** and **SB10** (channel 1), **SB6** and **SB8** (channel 2), **SB11** and **SB13** (channel 3), **SB9** and **SB12** (channel 4). These jumpers are highlighted in red. For each channel, both jumpers in the pair are closed by default, meaning **R** equals 24.95kΩ (see above). If both jumpers in the pair are opened, **R** for the corresponding channel becomes 49.9kΩ.

CONNECTING EXTERNAL SIGNALS

The **H1-H4** connectors on the module are used to connect external analog signals; it is recommended to insert removable 3.5mm pitch terminal blocks into them. The pin layout is marked on the **bottom-side** of the module. When **using unipolar input mode**, polarity must be observed when connecting external input signals.

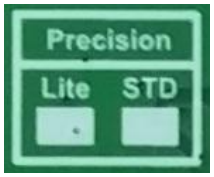
The following diagram shows the commutation for both voltage and current measurement. This commutation can be done for any channel, regardless of the commutation for other channels.



This commutation can be done for any channel, regardless of the commutation for other channels.

CONFIGURATION TABLES

Precision: Lite / STD. The module is available in two precision grades, differing only in the tolerance and temperature stability of the passive components in the measurement path (gain-setting resistors and filter/reference capacitors) — the circuit topology and component values are identical between grades.



- **STD**: ±0.1% thin-film resistors (±10ppm/°C) and C0G/NP0 capacitors (±5%) for maximum measurement accuracy and thermal stability.
- **Lite**: ±0.5–1% resistors (±25ppm/°C) and X7R capacitors (±10%) — a lower-cost option for applications where the highest precision is not required.



Information is provided on the **bottom-side** indicating whether the **Qwiic©** connectors for **I²C** are installed

ACCESSORIES:

The following accessories may be required for using the module:

- A set of four removable terminal blocks with a 3.5mm pitch for terminal blocks **H1-H4**
- A set of two standoffs and four screws for mounting the module into the **IoTbase** series base module.
- Cable for the **Qwiic©** connector